

Early Warning Risiko Ketidakberhasilan Mahasiswa dalam Mata Kuliah pada Minggu Keempat Menggunakan Supervised Learning dan Knowledge-Based Risk Layer pada Open University Learning Analytics Dataset

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Abstract. Hasil akhir mata kuliah memberi informasi penting bagi monitoring akademik dan perencanaan intervensi dini. Penelitian ini mengembangkan early warning risiko ketidakberhasilan mahasiswa dalam mata kuliah pada akhir minggu keempat menggunakan Open University Learning Analytics Dataset (OULAD). Setiap baris merepresentasikan satu mahasiswa pada satu module-presentation. Target dirumuskan sebagai klasifikasi biner, yaitu *AtRisk* untuk hasil akhir mata kuliah *Withdrawn* dan *Fail*, serta *Successful* untuk *Pass* dan *Distinction*. Fitur prediktor dibentuk dari data demografis, registrasi awal, assessment, dan aktivitas Virtual Learning Environment yang tersedia sampai hari ke-28. Evaluasi membandingkan Logistic Regression, Random Forest, dan XGBoost melalui hold-out test berbasis kelompok mahasiswa dan 5-fold GroupKFold. Random Forest dipilih berdasarkan recall *AtRisk* cross-validation tertinggi sebesar 0,7107. Pada hold-out test, model tersebut menghasilkan accuracy 0,7594, precision 0,8007, recall 0,7213, F1-score 0,7589, dan ROC-AUC 0,8396. Knowledge-based risk layer menggabungkan prediksi model dengan empat indikator perilaku awal untuk menghasilkan level *High Risk*, *Medium Risk*, dan *Low Risk* beserta alasan dan rekomendasi intervensi. Sistem gabungan meningkatkan recall menjadi 0,7866 dengan precision 0,7043. Hasil analitik disajikan melalui dashboard statis yang memuat indikator risiko, prioritas module-presentation, sinyal dominan, dan daftar mahasiswa untuk mendukung monitoring akademik. Pendekatan ini menyediakan integrasi prediksi, interpretasi berbasis aturan, dan visual decision support untuk intervensi dini pada tingkat mata kuliah.

Keywords. academic risk prediction, course withdrawal, course failure, supervised learning, OULAD, early warning system

I. INTRODUCTION

Kegagalan dan pengunduran diri dari mata kuliah menjadi perhatian dalam pendidikan tinggi karena berkaitan dengan capaian belajar, kualitas layanan akademik, dan efektivitas pengambilan keputusan institusional. Penelitian sebelumnya menunjukkan bahwa machine learning dapat memprediksi retensi mahasiswa menggunakan karakteristik sosiodemografis dan metrik engagement [1]. Early warning system memperluas manfaat prediksi dengan menempatkan identifikasi risiko pada periode ketika intervensi akademik masih dapat dilakukan [2]. Ketersediaan informasi risiko pada minggu-minggu awal memberi ruang bagi dosen, tutor, dan pengelola program studi untuk merancang tindak lanjut yang relevan.

Sistem pembelajaran digital menghasilkan data akademik dan jejak aktivitas yang dapat digunakan untuk membaca engagement mahasiswa. Data tersebut mencakup interaksi dengan Virtual Learning Environment (VLE), partisipasi assessment, performa penilaian, dan informasi registrasi awal. Studi mengenai digital traces memperlihatkan bahwa pola aktivitas Learning Management System dapat membantu memahami variasi performa dan risiko mahasiswa [3]. Open University Learning Analytics Dataset (OULAD) menyediakan data tersebut dalam struktur relasional yang mencakup profil mahasiswa, assessment, registrasi, dan lebih dari sepuluh juta catatan aktivitas VLE [4].

Nilai prediktif suatu model bergantung pada kesesuaian data dengan waktu keputusan. Penggunaan aktivitas seluruh semester atau status unregistration menghasilkan informasi yang kuat untuk klasifikasi retrospektif, sedangkan early

warning memerlukan fitur yang telah tersedia pada waktu prediksi. Penelitian ini menggunakan cut-off hari ke-28 untuk merepresentasikan akhir minggu keempat. Assessment dan aktivitas VLE setelah cut-off dikeluarkan dari fitur, sedangkan status unregistration diperlakukan sebagai informasi masa depan. Pembatasan temporal tersebut menghasilkan estimasi performa yang lebih dekat dengan kondisi intervensi dini.

Literatur menunjukkan bahwa kemampuan mendeteksi mahasiswa berisiko perlu dihubungkan dengan proses tindak lanjut agar memberi dampak institusional [2]. Model machine learning menghasilkan kelas dan probabilitas, sedangkan pemangku kepentingan memerlukan alasan risiko, prioritas, dan rekomendasi yang dapat dibaca secara operasional. Kebutuhan tersebut membentuk research gap pada integrasi antara evaluasi model, interpretasi berbasis aturan, dan penyajian visual untuk decision support.

Penelitian ini bertujuan mengembangkan klasifikasi biner risiko ketidakberhasilan mahasiswa dalam mata kuliah pada minggu keempat, mengevaluasi tiga algoritma supervised learning, dan mengintegrasikan model terpilih dengan knowledge-based risk layer. Hasil sistem diterjemahkan menjadi dashboard Business Intelligence untuk monitoring risiko dan prioritas intervensi akademik pada tingkat module-presentation.

Kontribusi penelitian terdiri atas empat bagian. Pertama, penelitian membentuk dataset early warning pada unit student-module-presentation dengan cut-off hari ke-28. Kedua, evaluasi menggunakan pemisahan berbasis *id_student* untuk menjaga independensi mahasiswa antara data pelatihan dan pengujian. Ketiga, knowledge-based risk layer menghasilkan level, alasan risiko, dan rekomendasi berdasarkan prediksi model serta perilaku awal. Keempat, dashboard mengubah keluaran analitik menjadi indikator yang mendukung keputusan pimpinan akademik, program studi, tutor, dosen wali, dan tim konseling.

II. RELATED WORKS

Penelitian student dropout dan academic performance analytics memanfaatkan data akademik, sosiodemografis, dan aktivitas digital untuk memprediksi keberlanjutan studi serta hasil pembelajaran. Informasi awal seperti karakteristik sosiodemografis, riwayat akademik, dan engagement digital memiliki nilai prediktif terhadap retensi mahasiswa [1]. Pada OULAD, ruang lingkup target berada pada hasil akhir mata kuliah untuk setiap student-module-presentation.

Early warning system mengarahkan hasil prediksi pada proses intervensi. Plak et al. menunjukkan bahwa informasi risiko perlu disertai rancangan tindak lanjut agar dapat memengaruhi hasil akademik [2]. Shou et al. menggunakan OULAD untuk memprediksi performa mahasiswa melalui multidimensional time-series yang menggabungkan learning behavior, assessment score, dan informasi demografis. Pada 20% durasi course, model MTAPSP harian mencapai accuracy 0,9179 dan F1-score 0,9180 untuk target biner *Pass* serta *Distinction* terhadap *Fail* serta *Withdrawn* [5]. Temuan tersebut menegaskan bahwa horizon pengamatan menjadi bagian penting dalam membaca performa early warning.

Penelitian berbasis OULAD terus berkembang melalui kombinasi fitur assessment, aktivitas VLE, dan profil mahasiswa. Jawad et al. menerapkan Random Forest dengan SMOTE dan memperoleh testing accuracy 0,892 pada skenario 260 hari [6]. Balabied dan Eid melaporkan accuracy serta F1-score 0,90 menggunakan Random Forest [7]. Ujkani et al. memperoleh accuracy 0,93 menggunakan custom neural network [8], sedangkan KANFormer dari Alnasyan et al. mencapai accuracy 0,9459 dan F1-score 0,9481 [9]. Shou et al. [5], Ujkani et al. [8], dan Alnasyan et al. [9] memakai definisi target yang sebanding. Perbedaan horizon, split, balancing, dan arsitektur model tetap menentukan konteks perbandingan.

Pendekatan hybrid mengintegrasikan beberapa sumber data untuk identifikasi mahasiswa berisiko [10]. Penelitian lain mencakup early warning berbasis XGBoost [11], AutoML [12], precision education [13], phased prediction [14], pola penggunaan LMS [15], dan analisis lintas institusi [16]. Keragaman tersebut menunjukkan bahwa fitur, horizon waktu, dan konteks institusi menentukan interpretasi performa model.

Penelitian ini mengambil posisi pada integrasi tiga komponen. Supervised learning digunakan untuk menghasilkan probabilitas risiko, knowledge-based risk layer menerjemahkan probabilitas dan sinyal perilaku menjadi alasan serta level prioritas, dan dashboard menyajikan hasil sebagai visual decision support. Cut-off minggu keempat menjaga kesesuaian temporal antara fitur dan waktu intervensi. Integrasi tersebut menghubungkan evaluasi teknis dengan kebutuhan monitoring akademik yang dapat ditindaklanjuti.

III. METHODOLOGY

A. Research Design

Penelitian menggunakan supervised binary classification untuk mendeteksi risiko ketidakberhasilan mahasiswa dalam mata kuliah pada akhir minggu keempat. Alurnya mencakup audit OULAD, pembentukan fitur dengan cut-off temporal, exploratory data analysis, pemisahan berbasis mahasiswa, evaluasi model, knowledge-based risk layer, dan dashboard Business Intelligence. Seluruh eksperimen menggunakan *random_state=42*.

B. Dataset and Unit of Analysis

Open University Learning Analytics Dataset (OULAD) [4] diperoleh melalui UCI Machine Learning Repository. Tabel yang digunakan meliputi *studentInfo*, *studentRegistration*, *assessments*, *studentAssessment*, *studentVle*, *courses*, dan *vle*. *studentInfo* memuat 32.593 baris dan *studentVle* memuat 10.655.280 catatan aktivitas.

Satu baris hasil preprocessing merepresentasikan seorang mahasiswa pada satu kombinasi *code_module* dan *code_presentation*. Unit student-module-presentation ini mengikuti struktur label *studentInfo* dan penggabungan data registrasi, assessment, serta VLE.

C. Target Label and Temporal Cut-off

Target dirumuskan sebagai klasifikasi biner pada tingkat mata kuliah. Label *AtRisk* diberikan kepada baris dengan *final_result* berupa *Withdrawn* atau *Fail*. Label *Successful* diberikan kepada baris dengan *final_result* berupa *Pass* atau *Distinction*. Status tersebut menjelaskan hasil mahasiswa pada suatu module-presentation. Dataset terdiri atas 17.208 baris *AtRisk* dan 15.385 baris *Successful*.

Horizon early warning ditetapkan pada hari ke-28. Submission assessment dipilih dengan kondisi *date_submitted* ≤ 28 , sedangkan aktivitas VLE dipilih dengan kondisi *date* ≤ 28 . Informasi *date_unregistration*, *has_unregistration*, hasil akhir, dan aktivitas setelah hari ke-28 dikeluarkan dari fitur prediktor. Hasil akhir hanya digunakan untuk membentuk target evaluasi.

D. Feature Construction

Fitur kategorikal mencakup *gender*, *region*, *highest_education*, *imd_band*, *age_band*, *disability*, *code_module*, dan *code_presentation*. Fitur numerik awal mencakup *num_of_prev_attempts*, *studied_credits*, dan *date_registration*.

Data *studentAssessment* dihubungkan dengan tabel *assessments* melalui *id_assessment* untuk memperoleh konteks module-presentation. Data sampai hari ke-28 diagregasi menjadi *assessment_count*, *assessment_score_mean*, *assessment_score_max*, dan *assessment_score_min*. Data *studentVle* sampai hari ke-28 diagregasi menjadi *vle_total_clicks*, *vle_active_days*, *vle_site_count*, dan *vle_last_activity_day*.

Nilai numerik dikonversi menggunakan *pd.to_numeric* dengan nilai yang tidak dapat dikonversi diperlakukan sebagai missing value. Agregat perilaku yang belum memiliki aktivitas diisi dengan nol. Pada pipeline model, missing value numerik diimputasi menggunakan median dan kemudian distandardisasi. Missing value kategorikal diimputasi menggunakan modus dan ditransformasikan dengan one-hot encoding. Seluruh transformasi dipelajari di dalam pipeline pada data pelatihan.

E. Data Splitting and Validation

Data dibagi menjadi 80% train-validation dan 20% hold-out test menggunakan *GroupShuffleSplit*. Variabel *id_student* digunakan sebagai grup sehingga setiap mahasiswa ditempatkan secara eksklusif pada salah satu bagian. Train-validation terdiri atas 26.122 baris dan test terdiri atas 6.471 baris. Pemeriksaan menunjukkan overlap mahasiswa sebesar nol.

Evaluasi cross-validation menggunakan 5-fold *GroupKFold* pada train-validation. Pemilihan model didasarkan pada rata-rata recall kelas *AtRisk*, kemudian F1-score *AtRisk* digunakan sebagai tie-breaker. Setelah model terpilih, performa akhir dilaporkan pada hold-out test.

F. Supervised Learning Models

Tiga algoritma dibandingkan. Logistic Regression digunakan sebagai baseline linear dengan *class_weight='balanced'*. Random Forest menggunakan 250

pohon dan *class_weight='balanced'*. XGBoost menggunakan 200 estimator, kedalaman maksimum 4, learning rate 0,08, subsample 0,9, dan colsample 0,9. Nilai *scale_pos_weight* dihitung dari distribusi kelas train-validation.

Metrik evaluasi mencakup accuracy, precision, recall, dan F1-score untuk kelas *AtRisk*, ROC-AUC, confusion matrix, serta classification report. Recall *AtRisk* menjadi kriteria utama karena early warning memprioritaskan cakupan mahasiswa berisiko yang dapat diarahkan pada proses verifikasi dan intervensi.

G. Knowledge-Based Risk Layer

Knowledge-based risk layer menggunakan empat indikator: *assessment_score_mean*, *assessment_count*, *vle_total_clicks*, dan *vle_active_days*. Threshold dihitung dari kuartil bawah data train-validation. Prosedur ini menjaga threshold tetap independen dari hold-out test. Hasil perhitungan menghasilkan threshold skor assessment 0, jumlah assessment 0, total klik VLE 47, dan hari aktif VLE 4.

Sinyal risiko diberikan ketika nilai indikator berada pada atau di bawah threshold. Status *High Risk* diberikan ketika model memprediksi *AtRisk* dan mahasiswa memiliki minimal dua sinyal aturan. Status *Medium Risk* diberikan ketika model memprediksi *AtRisk* atau mahasiswa memiliki minimal dua sinyal aturan. Status *Low Risk* diberikan pada kondisi lainnya.

Setiap baris menghasilkan probabilitas *AtRisk*, jumlah sinyal, alasan risiko, dan rekomendasi intervensi. Aktivitas VLE rendah diarahkan pada pengingat dan monitoring akses. Assessment rendah atau belum dikerjakan diarahkan pada pendampingan akademik. Kombinasi minimal tiga sinyal diarahkan pada konseling atau tindak lanjut dosen wali.

Evaluasi sistem gabungan memetakan *High Risk* dan *Medium Risk* sebagai *AtRisk*, lalu mengukur perubahan cakupan deteksi, precision, dan beban verifikasi.

H. Visual Analytics and Business Intelligence

Dashboard statis dibangun menggunakan Matplotlib dan Seaborn. Dashboard memuat KPI mahasiswa unik, jumlah dan persentase prioritas intervensi, distribusi level risiko, risiko per module-presentation, distribusi probabilitas, sinyal dominan, median aktivitas VLE dan assessment, confusion matrix, serta perbandingan model dengan knowledge layer.

Daftar prioritas diurutkan berdasarkan level risiko, probabilitas *AtRisk*, dan jumlah sinyal. Keluaran tersebut mendukung beberapa tingkat keputusan. Pimpinan akademik memperoleh gambaran skala risiko, program studi melihat konsentrasi risiko antar module-presentation, dan tutor atau dosen wali memperoleh alasan serta rekomendasi pada tingkat mahasiswa. Sistem diposisikan sebagai decision support yang membantu stakeholder menetapkan tindak lanjut berdasarkan bukti analitik dan pertimbangan akademik.

IV. RESULTS AND DISCUSSION

A. Dataset and Validation Results

Dataset hasil preprocessing memuat 32.593 student-module-presentation dari 28.785 mahasiswa unik. Kelas *AtRisk* berjumlah 17.208 baris dan kelas *Successful* berjumlah

15.385 baris. Train-validation memuat 26.122 baris, sedangkan hold-out test memuat 6.471 baris dengan 3.398 kasus *AtRisk* dan 3.073 kasus *Successful*. Pemisahan berbasis *id_student* menghasilkan overlap mahasiswa sebesar nol. Missing value terutama terdapat pada *imd_band* dan sebagian kecil *date_registration*; keduanya ditangani di dalam pipeline berdasarkan data pelatihan pada setiap fold.

B. Cross-Validation Performance

Tabel I menunjukkan hasil 5-fold GroupKFold. Random Forest menghasilkan mean recall *AtRisk* tertinggi sebesar 0,7107 dan dipilih sebagai model final sesuai kriteria seleksi. XGBoost menghasilkan accuracy, F1-score, dan ROC-AUC tertinggi, masing-masing sebesar 0,7584, 0,7522, dan 0,8440.

TABLE I
HASIL 5-FOLD GROUPKFOLD PADA TRAIN-VALIDATION

Metrik	Logistic Regression	Random Forest	XGBoost
Accuracy	0,7496 ± 0,0039	0,7524 ± 0,0013	0,7584 ± 0,0024
Precision <i>AtRisk</i>	0,8090 ± 0,0059	0,7989 ± 0,0136	0,8215 ± 0,0096
Recall <i>AtRisk</i>	0,6889 ± 0,0064	0,7107 ± 0,0085	0,6938 ± 0,0029
F1 <i>AtRisk</i>	0,7442 ± 0,0059	0,7521 ± 0,0030	0,7522 ± 0,0039
ROC-AUC	0,8330 ± 0,0027	0,8362 ± 0,0023	0,8440 ± 0,0019

C. Hold-Out Test Performance

Pada hold-out test, Random Forest mencapai accuracy 0,7594, precision *AtRisk* 0,8007, recall 0,7213, F1-score 0,7589, dan ROC-AUC 0,8396. XGBoost menghasilkan accuracy dan ROC-AUC tertinggi, sedangkan Random Forest mempertahankan recall *AtRisk* tertinggi. Dari 3.398 kasus *AtRisk*, Random Forest mengenali 2.451 kasus dan melewati 947 kasus.

TABLE II
PERFORMA MODEL PADA HOLD-OUT TEST

Metrik	Logistic Regression	Random Forest	XGBoost
Accuracy	0,7487	0,7594	0,7611
Precision <i>AtRisk</i>	0,8034	0,8007	0,8154
Recall <i>AtRisk</i>	0,6904	0,7213	0,7045
F1 <i>AtRisk</i>	0,7426	0,7589	0,7559
ROC-AUC	0,8311	0,8396	0,8443

Fig. 1 membandingkan metrik ketiga model, Fig. 2 menunjukkan confusion matrix Random Forest, dan Fig. 3 menyajikan kurva ROC. Ketiga model memiliki kurva ROC yang berdekatan. Pemilihan Random Forest mengikuti tujuan early warning yang menempatkan cakupan deteksi *AtRisk* sebagai prioritas model selection.

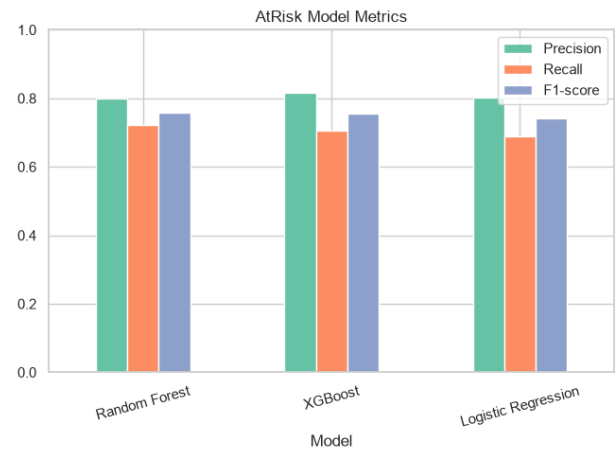


Fig. 1. Perbandingan metrik model pada hold-out test.

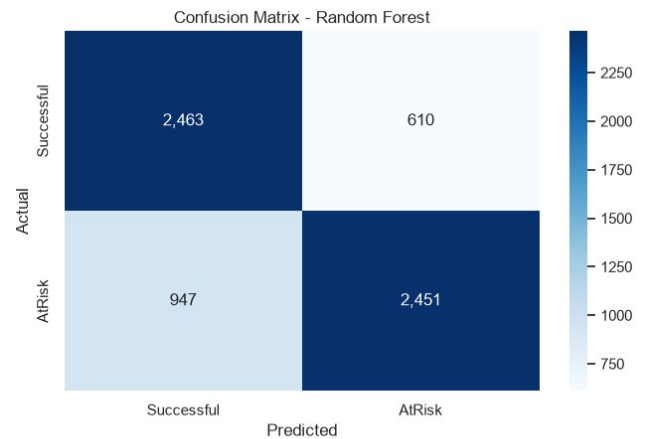


Fig. 2. Confusion matrix Random Forest pada hold-out test.

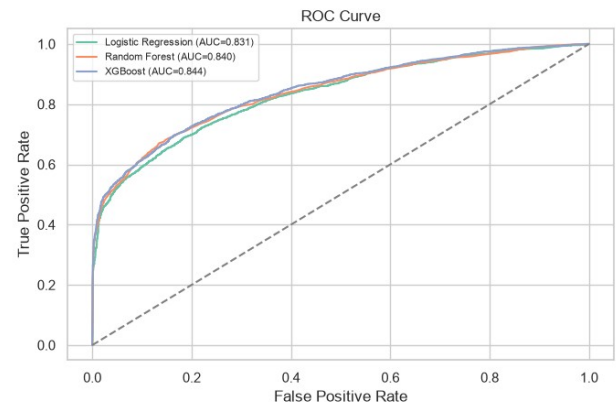


Fig. 3. Kurva ROC model pada hold-out test.

D. Feature Importance

Kontribusi fitur Random Forest didominasi sinyal perilaku awal. Total klik VLE, hari aktivitas terakhir, jumlah hari aktif, dan ragam situs yang diakses muncul sebagai prediktor teratas, diikuti fitur assessment dan registrasi. Nilai importance menunjukkan kontribusi prediktif global; interpretasi hubungan sebab akibat memerlukan desain penelitian kausal.

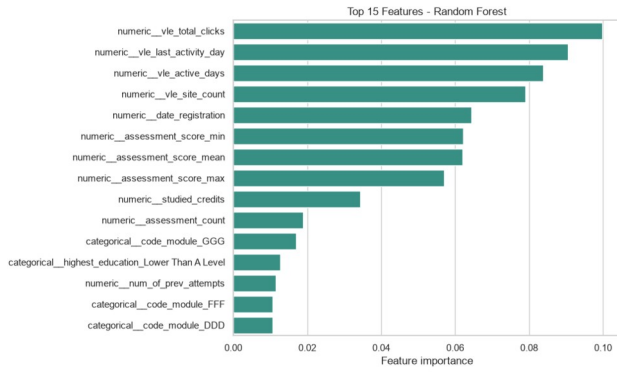


Fig. 4. Lima belas fitur dengan feature importance tertinggi pada Random Forest.

E. Benchmark with OULAD Studies

Fig. 5 menempatkan hasil penelitian dalam konteks lima studi OULAD. Shou et al. memakai target biner yang sama pada 20% durasi course [5]. Jawad et al. menggunakan data sampai 260 hari dan SMOTE [6], Balabied dan Eid menggunakan Random Forest [7], sedangkan Ujkani et al. [8] dan Alnasyan et al. [9] menggabungkan *Fail* serta *Withdrawn* sebagai kelompok at-risk. Grafik menyajikan nilai yang dilaporkan setiap studi sebagai benchmark kontekstual karena horizon, split, balancing, dan model yang digunakan berbeda.

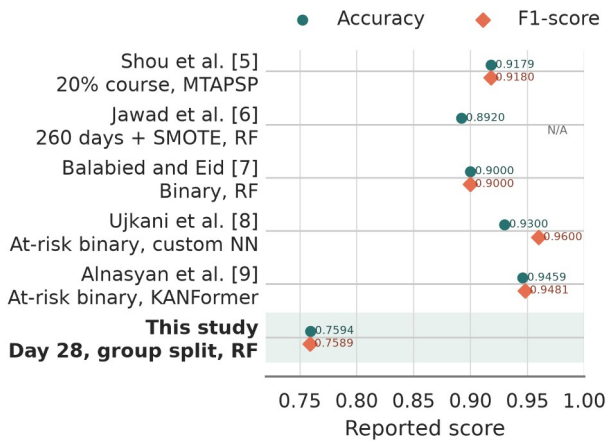


Fig. 5. Benchmark accuracy dan F1-score penelitian berbasis OULAD. Perbandingan bersifat kontekstual karena perbedaan horizon, data split, balancing, dan model.

F. Knowledge-Based Risk Layer and BI Output

Threshold kuartil bawah train-validation adalah skor assessment 0, jumlah assessment 0, total klik VLE 47, dan hari aktif VLE 4. Knowledge layer menghasilkan 1.816 *High Risk*, 1.979 *Medium Risk*, dan 2.676 *Low Risk*. Ketika *High Risk* serta *Medium Risk* dipetakan sebagai alarm *AtRisk*, recall meningkat dari 0,7213 menjadi 0,7866; precision berubah dari 0,8007 menjadi 0,7043. Perubahan tersebut memperluas cakupan alarm dan menambah kebutuhan verifikasi stakeholder.

TABLE III
PERBANDINGAN MODEL DAN KNOWLEDGE-BASED RISK LAYER

Metrik	Random Forest	RF + Knowledge Layer
Accuracy	0,7594	0,7146
Precision AtRisk	0,8007	0,7043
Recall AtRisk	0,7213	0,7866
F1 AtRisk	0,7589	0,7432

Dashboard pada Fig. 6 mengidentifikasi 3.795 student-module-presentation dalam antrean *High Risk* atau *Medium Risk*. Sinyal terbanyak adalah skor assessment rendah dengan 2.341 kasus. Daftar prioritas memuat identitas anonim, module-presentation, probabilitas *AtRisk*, level, jumlah sinyal, alasan, dan rekomendasi. Struktur ini menghubungkan evaluasi teknis dengan monitoring module-presentation dan tindak lanjut tingkat mahasiswa.

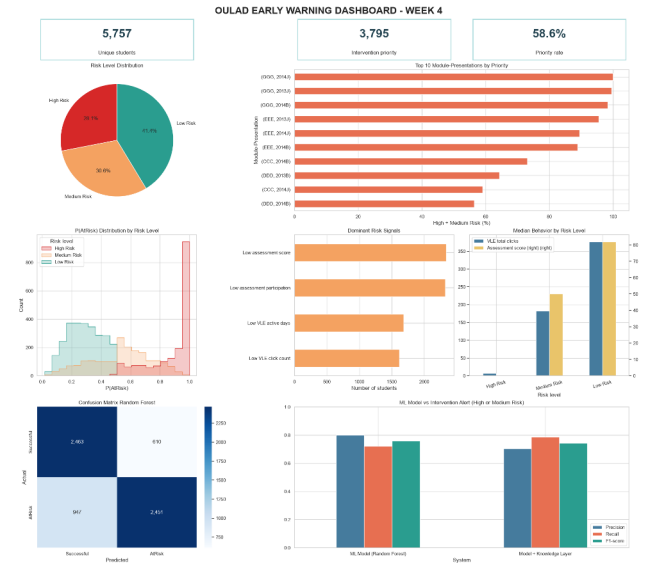


Fig. 6. Dashboard early warning OULAD pada akhir minggu keempat.

G. Discussion and Limitations

Hasil eksperimen memperlihatkan hubungan antara tujuan keputusan dan model selection. XGBoost unggul pada accuracy dan ROC-AUC, sementara Random Forest memperoleh recall *AtRisk* tertinggi pada cross-validation dan hold-out test. Performa sekitar 0,76 merepresentasikan skenario awal yang hanya menggunakan informasi sampai hari ke-28. Horizon ini menyediakan waktu intervensi lebih panjang sekaligus membatasi jumlah bukti perilaku yang tersedia bagi model.

Threshold assessment sebesar nol menunjukkan bahwa sebagian module-presentation belum memiliki submission sampai minggu keempat. Pengembangan berikutnya dapat menggunakan threshold per module-presentation atau menyesuaikan cut-off dengan jadwal assessment. Knowledge layer memperluas recall sebesar 0,0653 poin dan menghasilkan alasan operasional seperti aktivitas VLE rendah atau assessment yang belum dikerjakan. Alasan tersebut

menjadi bahan verifikasi bersama informasi kontekstual dari dosen dan tutor.

Dashboard memperluas fungsi model menjadi Business Intelligence melalui agregasi risiko, perbandingan module-presentation, dan antrean mahasiswa. Temuan 100% prioritas pada GGG 2014J menjadi sinyal untuk meninjau ukuran kelompok, jadwal assessment, serta karakteristik modul sebelum menentukan tindakan.

Keterbatasan penelitian mencakup konteks OULAD di Open University Inggris, fitur perilaku yang masih berupa agregat, konfigurasi model baseline, dan threshold kuartil yang memerlukan validasi pakar. Evaluasi mengukur performa deteksi, sedangkan dampak intervensi terhadap keberhasilan mata kuliah memerlukan penelitian lanjutan dengan data institusi dan desain evaluasi intervensi.

V. CONCLUSION

Penelitian ini mengembangkan early warning risiko ketidakberhasilan mahasiswa dalam mata kuliah pada akhir minggu keempat menggunakan OULAD. Dataset dibentuk pada unit student-module-presentation dengan fitur demografis, registrasi awal, assessment, dan aktivitas VLE sampai hari ke-28. Pemisahan berbasis *id_student* menjaga independensi mahasiswa antara train-validation dan hold-out test.

Random Forest dipilih berdasarkan recall *AtRisk* cross-validation tertinggi sebesar 0,7107. Pada hold-out test, model menghasilkan accuracy 0,7594, precision 0,8007, recall 0,7213, F1-score 0,7589, dan ROC-AUC 0,8396. Knowledge-based risk layer meningkatkan recall menjadi 0,7866 dengan menggabungkan prediksi model dan empat sinyal perilaku awal. Sistem menghasilkan level risiko, alasan, dan rekomendasi yang dapat diterjemahkan menjadi antrean intervensi.

Dashboard Business Intelligence menyajikan distribusi risiko, prioritas module-presentation, sinyal dominan, performa model, dan daftar mahasiswa anonim untuk monitoring. Integrasi supervised learning, knowledge-based risk layer, dan visual analytics menghasilkan decision support yang menghubungkan prediksi dengan proses tindak lanjut akademik.

Pengembangan berikutnya dapat membandingkan horizon minggu keempat, kedelapan, dan kedua belas; menggunakan threshold per module-presentation; melakukan tuning hyperparameter; serta menguji validitas eksternal pada data institusi lain. Evaluasi dampak intervensi juga diperlukan untuk mengukur kontribusi sistem terhadap keberhasilan mahasiswa menyelesaikan mata kuliah.

REFERENCES

- [1] S. Matz et al., "Using machine learning to predict student retention from socio-demographic characteristics and app-based engagement metrics," *Scientific Reports*, 2023, doi: 10.1038/s41598-023-32484-w.
- [2] S. Plak et al., "Early warning systems for more effective student counselling in higher education: Evidence from a Dutch field experiment," *Higher Education Quarterly*, 2022, doi: 10.1111/hequ.12298.
- [3] J. Pecuchova and M. Drlik, "Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces," *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3486762.
- [4] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," *Scientific Data*, vol. 4, article no. 170171, 2017, doi: 10.1038/sdata.2017.171.
- [5] Z. Shou, M. Xie, J. Mo, and H. Zhang, "Predicting Student Performance in Online Learning: A Multidimensional Time-Series Data Analysis Approach," *Applied Sciences*, vol. 14, no. 6, article no. 2522, 2024, doi: 10.3390/app14062522.
- [6] K. Jawad, M. A. Shah, and M. Tahir, "Students' Academic Performance and Engagement Prediction in a Virtual Learning Environment Using Random Forest with Data Balancing," *Sustainability*, vol. 14, no. 22, article no. 14795, 2022, doi: 10.3390/su142214795.
- [7] S. A. A. Balabied and H. F. Eid, "Utilizing Random Forest Algorithm for Early Detection of Academic Underperformance in Open Learning Environments," *PeerJ Computer Science*, vol. 9, article no. e1708, 2023, doi: 10.7717/peerj-cs.1708.
- [8] B. Ujkani, D. Minkovska, and N. Hinov, "Course Success Prediction and Early Identification of At-Risk Students Using Explainable Artificial Intelligence," *Electronics*, vol. 13, no. 21, article no. 4157, 2024, doi: 10.3390/electronics13214157.
- [9] B. Alnasyan, M. Basher, M. Allassafi, and K. Alnasyan, "Kanformer: An Attention-Enhanced Deep Learning Model for Predicting Student Performance in Virtual Learning Environments," *Social Network Analysis and Mining*, vol. 15, article no. 25, 2025, doi: 10.1007/s13278-025-01446-7.
- [10] T. Kustitskaya et al., "Hybrid Approach to Predicting Learning Success Based on Digital Educational History for Timely Identification of At-Risk Students," *Education Sciences*, 2024, doi: 10.3390/educsci14060657.
- [11] M. Carballo-Mendivil et al., "Predicting Student Dropout from Day One: XGBoost-Based Early Warning System Using Pre-Enrollment Data," *Applied Sciences*, 2025, doi: 10.3390/app15169202.
- [12] A. Garmpis et al., "Assisting Educational Analytics with AutoML Functionalities," *Computers*, 2022, doi: 10.3390/computers11060097.
- [13] C. Y. Tsai et al., "Precision education with statistical learning and deep learning: a case study in Taiwan," *International Journal of Educational Technology in Higher Education*, 2020, doi: 10.1186/s41239-020-00186-2.
- [14] M. V. Martins et al., "Multi-Class Phased Prediction of Academic Performance and Dropout in Higher Education," *Applied Sciences*, 2023, doi: 10.3390/app13084702.
- [15] J. R. Rico-Juan et al., "Study regarding the influence of a student's personality and an LMS usage profile on learning performance using machine learning techniques," *Applied Intelligence*, 2024, doi: 10.1007/s10489-024-05483-1.
- [16] J. Berens et al., "Crossing individual university boundaries: a comprehensive approach to predicting dropouts in the higher education system," *Higher Education*, 2025, doi: 10.1007/s10734-025-01509-w.

Early Warning of Students at Risk of Unsuccessful Course Outcomes in Week Four Using Supervised Learning and a Knowledge-Based Risk Layer on the Open University Learning Analytics Dataset

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Abstract. Final course outcomes provide important information for academic monitoring and early intervention planning. This study develops an early warning system for students at risk of unsuccessful course outcomes by the end of week four using the Open University Learning Analytics Dataset (OULAD). Each row represents one student in one module-presentation. The binary target is defined as *AtRisk* for *Withdrawn* and *Fail* outcomes and *Successful* for *Pass* and *Distinction*. Predictors are constructed from demographic data, initial registration, assessments, and Virtual Learning Environment activity available through day 28. The evaluation compares Logistic Regression, Random Forest, and XGBoost using a student-grouped hold-out test and 5-fold GroupKFold. Random Forest is selected because it achieves the highest cross-validation recall for *AtRisk* at 0.7107. On the hold-out test, the model achieves an accuracy of 0.7594, precision of 0.8007, recall of 0.7213, F1-score of 0.7589, and ROC-AUC of 0.8396. A knowledge-based risk layer combines model predictions with four early behavioral indicators to produce *High Risk*, *Medium Risk*, and *Low Risk* levels with reasons and intervention recommendations. The combined system increases recall to 0.7866 with a precision of 0.7043. The analytical results are presented through a static dashboard containing risk indicators, module-presentation priorities, dominant signals, and a student list for academic monitoring. This approach integrates prediction, rule-based interpretation, and visual decision support for early intervention at the course level.

Keywords. academic risk prediction, course withdrawal, course failure, supervised learning, OULAD, early warning system

I. INTRODUCTION

Course failure and withdrawal are important concerns in higher education because they are associated with learning outcomes, the quality of academic services, and the effectiveness of institutional decision-making. Previous

research has shown that machine learning can predict student retention from sociodemographic characteristics and engagement metrics [1]. Early warning systems extend the value of prediction by identifying risk while academic intervention remains feasible [2]. Risk information during the first weeks gives lecturers, tutors, and program administrators time to design relevant follow-up actions.

Digital learning systems generate academic data and activity traces that can be used to characterize student engagement. These data include interactions with the Virtual Learning Environment (VLE), assessment participation, assessment performance, and initial registration information. Research on digital traces has shown that Learning Management System activity patterns can help explain variations in student performance and risk [3]. The Open University Learning Analytics Dataset (OULAD) provides such data in a relational structure covering student profiles, assessments, registration, and more than ten million VLE activity records [4].

The predictive value of a model depends on whether its inputs are available at the time of decision. Full-semester activity or unregistration status may provide strong information for retrospective classification, whereas early warning requires features already observed at prediction time. This study uses a day-28 cut-off to represent the end of week four. Assessments and VLE activity after the cut-off are excluded from the predictors, and unregistration status is treated as future information. This temporal restriction yields a performance estimate that more closely represents an early-intervention setting.

The literature indicates that risk detection must be connected to follow-up processes to create institutional value [2]. Machine learning models produce classes and

probabilities, while stakeholders require risk reasons, priorities, and operationally interpretable recommendations. This need defines a research gap involving the integration of model evaluation, rule-based interpretation, and visual presentation for decision support.

This study aims to develop a binary classification of students at risk of unsuccessful course outcomes in week four, evaluate three supervised learning algorithms, and integrate the selected model with a knowledge-based risk layer. The system output is translated into a Business Intelligence dashboard for risk monitoring and intervention prioritization at the module-presentation level.

The study makes four contributions. First, it constructs an early warning dataset at the student-module-presentation level using a day-28 cut-off. Second, the evaluation uses an *id_student*-based split to maintain student independence between training and testing data. Third, the knowledge-based risk layer produces levels, reasons, and recommendations from model predictions and early behavior. Fourth, the dashboard transforms analytical outputs into indicators that support decisions by academic leaders, study programs, tutors, academic advisors, and counseling teams.

II. RELATED WORKS

Student dropout and academic performance analytics use academic, sociodemographic, and digital activity data to predict study continuation and learning outcomes. Early information, including sociodemographic characteristics, academic history, and digital engagement, has predictive value for student retention [1]. In OULAD, the target concerns the final course outcome for each student-module-presentation.

Early warning systems connect predictive results to intervention processes. Plak et al. showed that risk information should be accompanied by a follow-up design to influence academic outcomes [2]. Shou et al. used OULAD to predict student performance through a multidimensional time-series approach combining learning behavior, assessment scores, and demographic information. At 20% of the course duration, their daily MTAPSP model achieved an accuracy of 0.9179 and an F1-score of 0.9180 for a binary target contrasting *Pass* and *Distinction* with *Fail* and *Withdrawn* [5]. Their findings highlight the observation horizon as an important factor when interpreting early warning performance.

OULAD research has continued to combine assessment features, VLE activity, and student profiles. Jawad et al. applied Random Forest with SMOTE and obtained a testing accuracy of 0.892 in a 260-day scenario [6]. Balabied and Eid reported an accuracy and F1-score of 0.90 using Random Forest [7]. Ujkani et al. obtained an accuracy of 0.93 with a custom neural network [8], while the KANFormer proposed by Alnasyan et al. achieved an accuracy of 0.9459 and an F1-score of 0.9481 [9]. Shou et al. [5], Ujkani et al. [8], and Alnasyan et al. [9] used comparable target definitions. Differences in horizon, splitting, balancing, and model architecture still determine the context of comparison.

Hybrid approaches integrate multiple data sources to identify at-risk students [10]. Other studies cover XGBoost-

based early warning [11], AutoML [12], precision education [13], phased prediction [14], LMS usage patterns [15], and cross-institutional analysis [16]. This diversity shows that features, observation horizons, and institutional contexts shape the interpretation of model performance.

This study is positioned at the integration of three components. Supervised learning generates risk probabilities, a knowledge-based risk layer translates probabilities and behavioral signals into reasons and priority levels, and a dashboard presents the results as visual decision support. The week-four cut-off maintains temporal alignment between the predictors and the intervention point. This integration connects technical evaluation with actionable academic monitoring requirements.

III. METHODOLOGY

A. Research Design

This study uses supervised binary classification to identify students at risk of unsuccessful course outcomes by the end of week four. The workflow includes OULAD auditing, feature construction with a temporal cut-off, exploratory data analysis, student-grouped splitting, model evaluation, a knowledge-based risk layer, and a Business Intelligence dashboard. All experiments use *random_state=42*.

B. Dataset and Unit of Analysis

The Open University Learning Analytics Dataset (OULAD) [4] was obtained from the UCI Machine Learning Repository. The analysis uses the *studentInfo*, *studentRegistration*, *assessments*, *studentAssessment*, *studentVle*, *courses*, and *vle* tables. *studentInfo* contains 32,593 rows, while *studentVle* contains 10,655,280 activity records.

Each preprocessed row represents one student in one combination of *code_module* and *code_presentation*. This student-module-presentation unit follows the label structure in *studentInfo* and supports the consistent integration of registration, assessment, and VLE data.

C. Target Label and Temporal Cut-off

The target is formulated as a binary classification at the course level. The *AtRisk* label is assigned to rows whose *final_result* is *Withdrawn* or *Fail*. The *Successful* label is assigned to rows whose *final_result* is *Pass* or *Distinction*. These statuses describe a student's outcome in a specific module-presentation. The dataset contains 17,208 *AtRisk* rows and 15,385 *Successful* rows.

The early warning horizon is set at day 28. Assessment submissions are selected using *date_submitted* $\leq$ 28, and VLE activity is selected using *date* $\leq$ 28. The *date_unregistration*, *has_unregistration*, final outcome, and activity after day 28 are excluded from the predictors. The final outcome is used only to construct the evaluation target.

D. Feature Construction

Categorical features include *gender*, *region*, *highest_education*, *imd_band*, *age_band*, *disability*, *code_module*, and *code_presentation*. Initial numeric features

include `num_of_prev_attempts`, `studied_credits`, and `date_registration`.

The `studentAssessment` data are joined with `assessments` through `id_assessment` to obtain the module-presentation context. Data through day 28 are aggregated into `assessment_count`, `assessment_score_mean`, `assessment_score_max`, and `assessment_score_min`. The `studentVle` data through day 28 are aggregated into `vle_total_clicks`, `vle_active_days`, `vle_site_count`, and `vle_last_activity_day`.

Numeric values are converted with `pd.to_numeric`, and values that cannot be converted are treated as missing values. Behavioral aggregates with no recorded activity are filled with zero. Within the modeling pipeline, missing numeric values are imputed with the median and then standardized. Missing categorical values are imputed with the mode and transformed using one-hot encoding. All transformations are learned within the pipeline from the training data.

E. Data Splitting and Validation

The data are divided into 80% train-validation and 20% hold-out test sets using `GroupShuffleSplit`. The `id_student` variable is used as the group, ensuring that each student occurs exclusively in one partition. The train-validation set contains 26,122 rows and the test set contains 6,471 rows. The verification found zero student overlap.

Cross-validation uses 5-fold `GroupKFold` on the train-validation set. Model selection is based on the mean recall for the *AtRisk* class, with the *AtRisk* F1-score used as a tie-breaker. Final performance is then reported on the hold-out test.

F. Supervised Learning Models

Three algorithms are compared. Logistic Regression is used as a linear baseline with `class_weight='balanced'`. Random Forest uses 250 trees and `class_weight='balanced'`. XGBoost uses 200 estimators, a maximum depth of 4, a learning rate of 0.08, a subsample ratio of 0.9, and a colsample ratio of 0.9. The `scale_pos_weight` value is calculated from the train-validation class distribution.

Evaluation metrics include accuracy, precision, recall, and F1-score for the *AtRisk* class, ROC-AUC, a confusion matrix, and a classification report. *AtRisk* recall is the primary criterion because the early warning setting prioritizes the coverage of at-risk students who can be directed to verification and intervention processes.

G. Knowledge-Based Risk Layer

The knowledge-based risk layer uses four indicators: `assessment_score_mean`, `assessment_count`, `vle_total_clicks`, and `vle_active_days`. Thresholds are calculated from the lower quartile of the train-validation data, keeping them independent of the hold-out test. The calculation produces thresholds of 0 for assessment score, 0 for assessment count, 47 for total VLE clicks, and 4 for active VLE days.

A risk signal is activated when an indicator is at or below its threshold. *High Risk* is assigned when the model predicts *AtRisk* and the student has at least two rule signals. *Medium*

Risk is assigned when the model predicts *AtRisk* or the student has at least two rule signals. *Low Risk* is assigned otherwise.

Each row produces an *AtRisk* probability, a signal count, risk reasons, and an intervention recommendation. Low VLE activity leads to access reminders and monitoring. Low or missing assessment participation leads to academic support. A combination of at least three signals leads to counseling or follow-up by an academic advisor.

The combined system is evaluated by mapping *High Risk* and *Medium Risk* to *AtRisk* and measuring changes in detection coverage, precision, and verification workload.

H. Visual Analytics and Business Intelligence

The static dashboard is constructed with Matplotlib and Seaborn. It presents KPIs for unique students, the number and percentage prioritized for intervention, the risk-level distribution, risk by module-presentation, the probability distribution, dominant signals, median VLE and assessment activity, a confusion matrix, and a comparison between the model and the knowledge layer.

The priority list is sorted by risk level, *AtRisk* probability, and signal count. The output supports decisions at several levels. Academic leaders obtain an overview of risk scale, study programs see concentrations of risk across module-presentations, and tutors or academic advisors receive student-level reasons and recommendations. The system serves as decision support for stakeholder follow-up based on analytical evidence and academic judgment.

IV. RESULTS AND DISCUSSION

A. Dataset and Validation Results

The preprocessed dataset contains 32,593 student-module-presentations from 28,785 unique students. The *AtRisk* class contains 17,208 rows and the *Successful* class contains 15,385 rows. The train-validation set contains 26,122 rows, while the hold-out test contains 6,471 rows with 3,398 *AtRisk* and 3,073 *Successful* cases. The `id_student`-based split produces zero student overlap. Missing values occur mainly in `imd_band` and in a small proportion of `date_registration`; both are handled within the pipeline using the training data in each fold.

B. Cross-Validation Performance

Table I presents the 5-fold GroupKFold results. Random Forest achieves the highest mean *AtRisk* recall at 0.7107 and is selected as the final model according to the selection criterion. XGBoost achieves the highest accuracy, F1-score, and ROC-AUC at 0.7584, 0.7522, and 0.8440, respectively.

TABLE I
5-FOLD GROUPKFOLD RESULTS ON THE TRAIN-VALIDATION SET

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	0.7496 ± 0.0039	0.7524 ± 0.0013	0.7584 ± 0.0024
AtRisk Precision	0.8090 ± 0.0059	0.7989 ± 0.0136	0.8215 ± 0.0096
AtRisk Recall	0.6889 ± 0.0064	0.7107 ± 0.0085	0.6938 ± 0.0029
AtRisk F1	0.7442 ± 0.0059	0.7521 ± 0.0030	0.7522 ± 0.0039
ROC-AUC	0.8330 ± 0.0027	0.8362 ± 0.0023	0.8440 ± 0.0019

C. Hold-Out Test Performance

On the hold-out test, Random Forest achieves an accuracy of 0.7594, *AtRisk* precision of 0.8007, recall of 0.7213, F1-score of 0.7589, and ROC-AUC of 0.8396. XGBoost achieves the highest accuracy and ROC-AUC, while Random Forest maintains the highest *AtRisk* recall. Of the 3,398 *AtRisk* cases, Random Forest identifies 2,451 and misses 947.

TABLE II
MODEL PERFORMANCE ON THE HOLD-OUT TEST

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	0.7487	0.7594	0.7611
<i>AtRisk</i> Precision	0.8034	0.8007	0.8154
<i>AtRisk</i> Recall	0.6904	0.7213	0.7045
<i>AtRisk</i> F1	0.7426	0.7589	0.7559
ROC-AUC	0.8311	0.8396	0.8443

Fig. 1 compares the metrics of the three models, Fig. 2 shows the Random Forest confusion matrix, and Fig. 3 presents the ROC curves. The three ROC curves are close. Random Forest is selected because the early warning objective prioritizes *AtRisk* detection coverage.

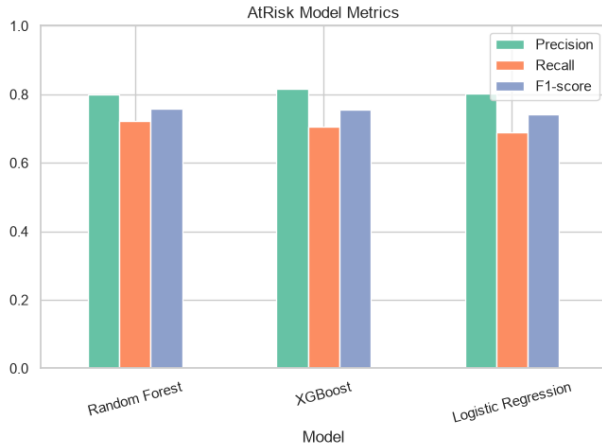


Fig. 1. Model metric comparison on the hold-out test.

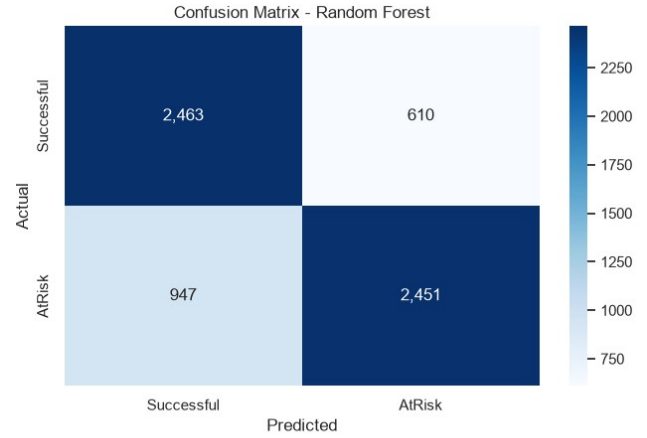


Fig. 2. Random Forest confusion matrix on the hold-out test.

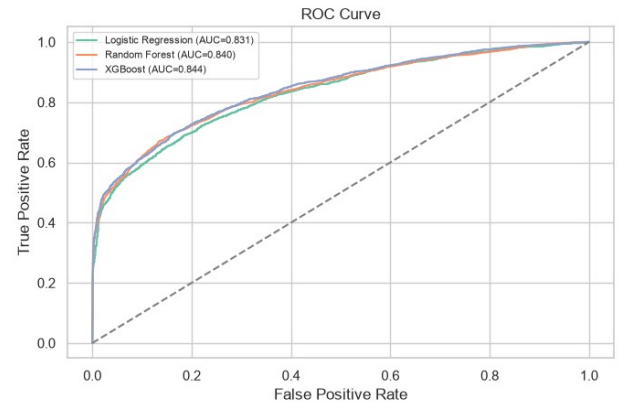


Fig. 3. Model ROC curves on the hold-out test.

D. Feature Importance

Early behavioral signals dominate the Random Forest feature contributions. Total VLE clicks, last activity day, active days, and the number of accessed sites appear as the leading predictors, followed by assessment and registration features. Importance values indicate global predictive contributions; causal interpretation requires a causal research design.

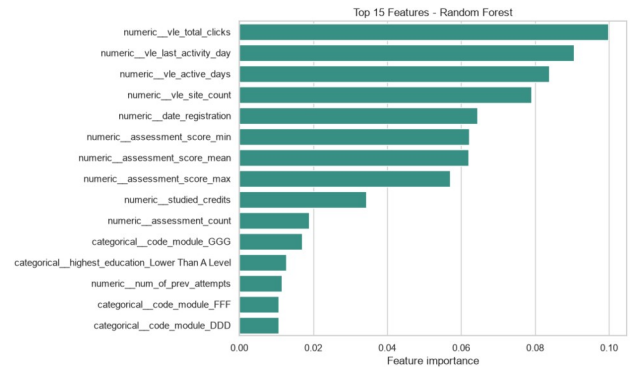


Fig. 4. Fifteen features with the highest Random Forest feature importance.

E. Benchmark with OULAD Studies

Fig. 5 positions the results in the context of five OULAD studies. Shou et al. used the same binary target at 20% of course duration [5]. Jawad et al. used data through day 260 and SMOTE [6], Balabied and Eid used Random Forest [7], and Ujkani et al. [8] and Alnasyan et al. [9] combined *Fail* and *Withdrawn* as the at-risk group. The figure presents the reported values as a contextual benchmark because the studies differ in horizon, data split, balancing, and model.

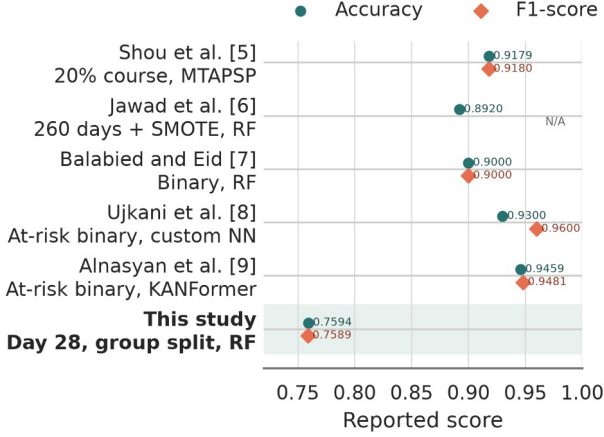


Fig. 5. Accuracy and F1-score benchmark for OULAD-based studies. The comparison is contextual because the studies use different horizons, data splits, balancing methods, and models.

F. Knowledge-Based Risk Layer and BI Output

The lower-quartile thresholds from the train-validation set are an assessment score of 0, an assessment count of 0, 47 total VLE clicks, and 4 active VLE days. The knowledge layer produces 1,816 *High Risk*, 1,979 *Medium Risk*, and 2,676 *Low Risk* cases. When *High Risk* and *Medium Risk* are mapped to an *AtRisk* alert, recall increases from 0.7213 to 0.7866, while precision changes from 0.8007 to 0.7043. This change expands alert coverage and increases the stakeholder verification workload.

TABLE III
MODEL AND KNOWLEDGE-BASED RISK LAYER COMPARISON

Metric	Random Forest	RF + Knowledge Layer
Accuracy	0.7594	0.7146
AtRisk Precision	0.8007	0.7043
AtRisk Recall	0.7213	0.7866
AtRisk F1	0.7589	0.7432

The dashboard in Fig. 6 identifies 3,795 student-module-presentations in the *High Risk* or *Medium Risk* queue. The most frequent signal is low assessment score, with 2,341 cases. The priority list contains anonymous identities, module-presentations, *AtRisk* probabilities, levels, signal counts, reasons, and recommendations. This structure connects technical evaluation with module-presentation monitoring and student-level follow-up.

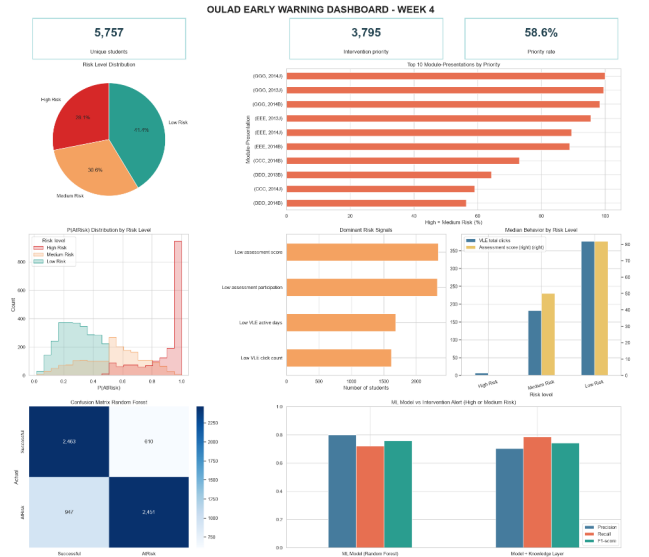


Fig. 6. OULAD early warning dashboard at the end of week four.

G. Discussion and Limitations

The experimental results show the relationship between the decision objective and model selection. XGBoost performs best in accuracy and ROC-AUC, while Random Forest achieves the highest *AtRisk* recall in both cross-validation and the hold-out test. Performance near 0.76 represents an early scenario that uses only information available through day 28. This horizon provides more time for intervention while limiting the behavioral evidence available to the model.

The assessment threshold of zero indicates that some module-presentations have no submissions by week four. Future work can use module-presentation-specific thresholds or align the cut-off with assessment schedules. The knowledge layer increases recall by 0.0653 points and produces operational reasons such as low VLE activity or incomplete assessments. These reasons support verification using contextual information from lecturers and tutors.

The dashboard extends the model into Business Intelligence through risk aggregation, module-presentation comparison, and a student queue. The finding that 100% of GGG 2014J cases are prioritized signals the need to review group size, assessment schedules, and module characteristics before selecting an action.

The study is limited by the context of OULAD at the United Kingdom Open University, behavioral features represented as aggregates, baseline model configurations, and quartile thresholds that require expert validation. The evaluation measures detection performance, while the impact of interventions on successful course outcomes requires further research using institutional data and an intervention evaluation design.

V. CONCLUSION

This study develops an early warning system for students at risk of unsuccessful course outcomes by the end of week four using OULAD. The dataset is constructed at the student-module-presentation level using demographic, initial

registration, assessment, and VLE activity features available through day 28. The *id_student*-based split maintains student independence between the train-validation and hold-out test sets.

Random Forest is selected based on the highest cross-validation *AtRisk* recall of 0.7107. On the hold-out test, the model achieves an accuracy of 0.7594, precision of 0.8007, recall of 0.7213, F1-score of 0.7589, and ROC-AUC of 0.8396. The knowledge-based risk layer increases recall to 0.7866 by combining the model prediction with four early behavioral signals. The system produces risk levels, reasons, and recommendations that can be translated into an intervention queue.

The Business Intelligence dashboard presents the risk distribution, module-presentation priorities, dominant signals, model performance, and an anonymous student list for monitoring. The integration of supervised learning, a knowledge-based risk layer, and visual analytics produces decision support that connects prediction with academic follow-up processes.

Future work can compare week-four, week-eight, and week-twelve horizons; use module-presentation-specific thresholds; tune hyperparameters; and test external validity on other institutional datasets. Intervention impact evaluation is also required to measure the system's contribution to successful course completion.

REFERENCES

- [1] S. Matz et al., "Using machine learning to predict student retention from socio-demographic characteristics and app-based engagement metrics," *Scientific Reports*, 2023, doi: 10.1038/s41598-023-32484-w.
- [2] S. Plak et al., "Early warning systems for more effective student counselling in higher education: Evidence from a Dutch field experiment," *Higher Education Quarterly*, 2022, doi: 10.1111/hequ.12298.
- [3] J. Pecuchova and M. Drlik, "Enhancing the Early Student Dropout Prediction Model Through Clustering Analysis of Students' Digital Traces," *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3486762.
- [4] J. Kuzilek, M. Hlostá, and Z. Zdrahal, "Open University Learning Analytics dataset," *Scientific Data*, vol. 4, article no. 170171, 2017, doi: 10.1038/sdata.2017.171.
- [5] Z. Shou, M. Xie, J. Mo, and H. Zhang, "Predicting Student Performance in Online Learning: A Multidimensional Time-Series Data Analysis Approach," *Applied Sciences*, vol. 14, no. 6, article no. 2522, 2024, doi: 10.3390/app14062522.
- [6] K. Jawad, M. A. Shah, and M. Tahir, "Students' Academic Performance and Engagement Prediction in a Virtual Learning Environment Using Random Forest with Data Balancing," *Sustainability*, vol. 14, no. 22, article no. 14795, 2022, doi: 10.3390/su142214795.
- [7] S. A. A. Balabied and H. F. Eid, "Utilizing Random Forest Algorithm for Early Detection of Academic Underperformance in Open Learning Environments," *PeerJ Computer Science*, vol. 9, article no. e1708, 2023, doi: 10.7717/peerj-cs.1708.
- [8] B. Ujkani, D. Minkovska, and N. Hinov, "Course Success Prediction and Early Identification of At-Risk Students Using Explainable Artificial Intelligence," *Electronics*, vol. 13, no. 21, article no. 4157, 2024, doi: 10.3390/electronics13214157.
- [9] B. Alnasyan, M. Basher, M. Alassafi, and K. Alnasyan, "Kanformer: An Attention-Enhanced Deep Learning Model for Predicting Student Performance in Virtual Learning Environments," *Social Network Analysis and Mining*, vol. 15, article no. 25, 2025, doi: 10.1007/s13278-025-01446-7.
- [10] T. Kustitskaya et al., "Hybrid Approach to Predicting Learning Success Based on Digital Educational History for Timely Identification of At-Risk Students," *Education Sciences*, 2024, doi: 10.3390/educsci14060657.
- [11] M. Carballo-Mendivil et al., "Predicting Student Dropout from Day One: XGBoost-Based Early Warning System Using Pre-Enrollment Data," *Applied Sciences*, 2025, doi: 10.3390/app15169202.
- [12] A. Garmpis et al., "Assisting Educational Analytics with AutoML Functionalities," *Computers*, 2022, doi: 10.3390/computers11060097.
- [13] C. Y. Tsai et al., "Precision education with statistical learning and deep learning: a case study in Taiwan," *International Journal of Educational Technology in Higher Education*, 2020, doi: 10.1186/s41239-020-00186-2.
- [14] M. V. Martins et al., "Multi-Class Phased Prediction of Academic Performance and Dropout in Higher Education," *Applied Sciences*, 2023, doi: 10.3390/app13084702.
- [15] J. R. Rico-Juan et al., "Study regarding the influence of a student's personality and an LMS usage profile on learning performance using machine learning techniques," *Applied Intelligence*, 2024, doi: 10.1007/s10489-024-05483-1.
- [16] J. Berens et al., "Crossing individual university boundaries: a comprehensive approach to predicting dropouts in the higher education system," *Higher Education*, 2025, doi: 10.1007/s10734-025-01509-w.